

DISCOVERY ISLAND, BC, Canada

WMO: 710310

Lat: 48.4247N

Lon: 123.2258W

Elev: 62

StdP: 14.66

Time Zone: -8.00 (INT)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	30.0	33.9	12.0	10.2	32.8	16.1	12.5	35.0	47.0	46.0	43.0	46.6	18.0	40	0.596

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.6	72.8	60.4	69.2	58.7	66.3	57.7	61.0	69.2	59.4	66.1	58.0	63.6	7.5	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.5	70.7	62.6	56.3	67.6	61.3	55.2	65.0	60.2	27.1	69.7	26.0	66.4	25.1	63.8	65.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 35.9	28.2	21.8	DB	26.8	81.1	5.2	4.3	23.1	84.2	20.1	86.7	17.2	89.1	13.4	92.2	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.3	44.1	44.2	46.2	49.5	53.6	57.0	59.3	59.8	57.7	52.3	47.5	44.0		
	DBStd	6.98	4.48	4.30	3.74	3.32	3.55	3.46	3.55	3.36	3.35	2.95	4.11	4.48		
	HDD50	821	187	163	126	47	6	0	0	0	0	11	91	190		
	HDD65	5021	649	582	584	465	353	245	183	167	222	394	525	652		
	CDD50	1293	3	1	7	31	119	209	288	305	230	83	15	2		
	CDD65	19	0	0	0	0	0	4	7	6	2	0	0	0		
	CDH74	79	0	0	0	0	0	16	34	27	2	0	0	0		
	CDH80	9	0	0	0	0	0	1	6	2	0	0	0	0		
	WSAvg	9.2	12.2	11.6	9.9	8.1	7.1	7.2	7.5	7.1	6.8	8.4	11.3	13.2		
	PrecAvg	25.4	4.2	2.9	1.9	1.3	1.0	0.7	0.6	0.8	1.3	2.4	3.9	4.4		
(15) Precipitation	PrecMax	34.4	8.5	6.7	4.3	2.8	2.0	2.7	1.5	2.6	2.8	8.1	7.5	10.4		
	PrecMin	15.3	0.6	0.4	0.5	0.7	0.2	0.1	0.0	0.0	0.2	0.2	0.6	0.4		
	PrecStd	4.9	2.1	1.5	1.1	0.6	0.5	0.6	0.4	0.7	0.7	1.8	1.7	2.1		
	PrecStd	4.9	2.1	1.5	1.1	0.6	0.5	0.6	0.4	0.7	0.7	1.8	1.7	2.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.4	53.4	56.5	63.6	69.3	76.1	78.5	77.7	72.8	62.5	56.7	54.6		
		MCWB	51.3	49.0	49.1	51.8	54.3	60.0	60.8	60.6	60.9	56.8	52.7	49.9		
	2%	DB	52.0	51.1	53.7	58.4	64.5	69.9	71.8	72.3	68.4	59.5	54.6	52.1		
		MCWB	47.9	48.3	48.4	49.2	53.2	59.0	59.0	60.9	57.7	52.8	51.7	48.5		
	5%	DB	50.4	49.9	52.0	55.6	61.0	64.7	67.5	68.4	65.5	57.6	53.3	50.4		
		MCWB	47.5	46.9	47.7	48.1	52.9	56.7	58.7	59.6	56.4	52.0	50.8	47.3		
	10%	DB	49.2	48.8	50.6	53.7	58.4	61.9	64.5	65.4	63.1	56.0	52.2	49.1		
		MCWB	46.8	46.1	46.8	47.5	52.5	55.0	57.5	58.5	56.0	51.5	49.9	46.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.5	50.0	51.1	51.8	56.6	60.7	61.1	63.3	62.5	57.1	53.7	50.6		
		MCDB	50.2	51.5	53.7	56.5	65.3	68.1	68.1	73.7	71.5	59.5	55.1	52.7		
	2%	WB	47.7	48.4	49.0	49.8	54.7	57.6	59.2	61.6	59.7	54.2	52.2	48.0		
		MCDB	49.0	49.8	51.5	53.6	61.1	64.0	65.7	71.0	65.0	56.1	53.9	49.6		
	5%	WB	46.7	47.0	47.7	48.5	53.5	56.0	58.0	60.1	57.8	52.6	50.9	46.7		
		MCDB	48.3	48.4	50.5	51.6	58.5	61.4	63.9	68.3	61.7	54.6	52.5	48.3		
	10%	WB	45.5	45.5	46.8	47.6	52.4	54.8	56.9	58.5	56.5	51.5	49.6	45.5		
		MCDB	47.1	47.1	49.2	50.6	56.8	59.4	62.0	64.8	59.7	53.6	51.0	47.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.4	4.8	6.1	7.4	8.6	9.6	10.4	10.6	9.0	5.3	4.5	4.3		
		MCDBR	5.5	5.4	7.4	9.8	13.2	14.6	15.5	15.8	13.2	7.4	5.4	5.7		
		MCWBR	4.5	3.8	4.4	5.3	6.4	7.4	7.2	8.3	6.6	4.6	4.7	4.8		
		MCDBR	5.2	4.9	7.0	7.6	11.1	12.1	12.4	15.7	11.2	6.8	5.6	5.1		
	5% WB	MCWBR	4.7	3.8	4.3	4.8	6.0	6.8	6.6	8.2	6.7	4.6	4.8	4.4		
		MCWBR	4.7	3.8	4.3	4.8	6.0	6.8	6.6	8.2	6.7	4.6	4.8	4.4		
(40) Clear-Sky Solar Irradiance	taub		0.320	0.320	0.333	0.359	0.380	0.357	0.350	0.355	0.341	0.333	0.327	0.316		
	taud		2.521	2.543	2.521	2.439	2.398	2.481	2.536	2.520	2.559	2.562	2.501	2.483		
	Ebn at Noon		240	269	282	283	280	287	287	281	276	259	233	224		
	Edh at Noon		19	23	28	33	36	33	31	30	27	23	20	18		
(44) All-Sky Solar Radiation	RadAvg		296	576	899	1360	1737	1900	2076	1752	1249	688	354	238		
	RadStd		39	77	105	127	137	180	145	113	102	75	54	31		

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
(47)	Regional (75 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page